

## SELFRIDGE AFB, MI, USA

WMO: 725377

Lat: 42.608N

Lon: 82.818W

Elev: 177

StdP: 99.22

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14804

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -17.5      | -14.8 | -21.3                         | 0.6 | -16.4 | -18.4 | 0.8 | -13.8 | 11.4                  | -4.4 | 10.5 | -3.8 | 3.8                   | 230  | 0.536 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.9                   | 32.2            | 23.4 | 30.3 | 22.4 | 28.9 | 21.7 | 24.9                | 29.5 | 23.9 | 28.2 | 23.0 | 27.1 | 5.2                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.7                            | 18.9 | 27.1 | 22.6 | 17.7 | 26.3 | 21.8 | 16.9 | 25.7 | 76.4          | 29.5 | 72.6 | 28.2 | 69.1 | 27.1 | 29.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| Min               | Max  |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 9.8               | 8.7  | 7.9 | -21.1                      | 34.9 | 3.7                | 1.5 | -23.8                                              | 36.0 | -25.9      | 36.9 | -28.0      | 37.7 | -30.7      | 38.8 |  |  |
|                   |      |     | -21.3                      | 26.8 | 3.7                | 1.3 | -24.0                                              | 27.8 | -26.2      | 28.6 | -28.2      | 29.3 | -30.9      | 30.3 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 9.6         | -4.3  | -2.4  | 1.4   | 7.9   | 14.5  | 20.0  | 22.3  | 21.6  | 18.0  | 11.5  | 4.9   | -0.6  | (6)  |
|                                                                        | DBStd       | 10.48       | 6.17  | 6.02  | 5.73  | 4.85  | 4.53  | 3.93  | 3.17  | 2.97  | 4.10  | 4.51  | 4.79  | 5.35  | (7)  |
|                                                                        | HDD10.0     | 1702        | 443   | 348   | 274   | 95    | 10    | 0     | 0     | 0     | 1     | 36    | 166   | 329   | (8)  |
|                                                                        | HDD18.3     | 3561        | 702   | 581   | 526   | 314   | 138   | 27    | 4     | 6     | 56    | 218   | 403   | 586   | (9)  |
|                                                                        | CDD10.0     | 1566        | 1     | 1     | 7     | 32    | 148   | 300   | 382   | 360   | 241   | 81    | 12    | 2     | (10) |
|                                                                        | CDD18.3     | 383         | 0     | 0     | 1     | 2     | 18    | 77    | 128   | 108   | 46    | 5     | 0     | 0     | (11) |
|                                                                        | CDH23.3     | 3238        | 0     | 0     | 3     | 24    | 184   | 713   | 1127  | 809   | 345   | 32    | 0     | 0     | (12) |
|                                                                        | CDH26.7     | 963         | 0     | 0     | 0     | 4     | 42    | 246   | 363   | 218   | 87    | 4     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 3.8         | 4.6   | 4.4   | 4.5   | 4.4   | 3.7   | 3.2   | 3.1   | 3.0   | 3.1   | 3.7   | 4.2   | 4.2   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 848         | 55    | 47    | 62    | 82    | 92    | 77    | 85    | 82    | 80    | 62    | 63    | 61    | (15) |
|                                                                        | PrecMax     | 1201        | 104   | 114   | 133   | 162   | 184   | 137   | 172   | 159   | 171   | 168   | 141   | 102   | (16) |
|                                                                        | PrecMin     | 595         | 15    | 7     | 23    | 19    | 38    | 27    | 17    | 14    | 13    | 12    | 18    | 22    | (17) |
|                                                                        | PrecStd     | 126         | 24    | 26    | 28    | 33    | 40    | 34    | 44    | 34    | 46    | 32    | 29    | 21    | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 12.8  | 16.2  | 22.3  | 26.4  | 30.0  | 33.4  | 34.0  | 33.0  | 32.1  | 27.0  | 19.1  | 14.9  | (19) |
|                                                                        |             | MCWB        | 11.5  | 11.9  | 14.3  | 17.4  | 20.6  | 23.5  | 24.7  | 24.7  | 23.2  | 20.7  | 14.7  | 12.8  | (20) |
|                                                                        | 2%          | DB          | 9.7   | 11.8  | 16.8  | 22.5  | 27.7  | 31.7  | 32.1  | 30.9  | 28.9  | 23.6  | 16.6  | 11.4  | (21) |
|                                                                        |             | MCWB        | 8.0   | 9.1   | 12.0  | 15.3  | 20.0  | 22.8  | 23.6  | 23.0  | 21.6  | 18.4  | 13.6  | 9.8   | (22) |
|                                                                        | 5%          | DB          | 5.9   | 7.9   | 12.9  | 19.1  | 25.4  | 29.8  | 30.2  | 29.0  | 27.3  | 21.3  | 14.5  | 8.8   | (23) |
|                                                                        |             | MCWB        | 4.4   | 5.5   | 9.3   | 13.2  | 18.8  | 21.9  | 22.6  | 22.4  | 20.5  | 17.2  | 12.0  | 7.0   | (24) |
|                                                                        | 10%         | DB          | 3.5   | 5.1   | 9.8   | 16.3  | 22.8  | 27.5  | 28.7  | 27.5  | 25.1  | 19.0  | 12.4  | 6.3   | (25) |
|                                                                        |             | MCWB        | 2.4   | 3.2   | 6.6   | 11.0  | 17.3  | 20.7  | 21.8  | 21.3  | 19.6  | 15.6  | 10.0  | 4.6   | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 12.1  | 13.0  | 15.4  | 19.2  | 23.0  | 25.6  | 26.6  | 25.7  | 24.4  | 21.9  | 16.3  | 13.0  | (27) |
|                                                                        |             | MCDB        | 12.7  | 14.4  | 20.0  | 25.1  | 26.7  | 30.7  | 31.5  | 30.9  | 29.0  | 25.4  | 17.5  | 13.9  | (28) |
|                                                                        | 2%          | WB          | 8.0   | 9.2   | 12.6  | 16.6  | 21.5  | 24.0  | 25.1  | 24.5  | 23.1  | 19.9  | 14.3  | 10.2  | (29) |
|                                                                        |             | MCDB        | 8.8   | 11.3  | 15.9  | 20.7  | 25.8  | 29.7  | 29.6  | 28.5  | 27.2  | 22.1  | 15.9  | 11.4  | (30) |
|                                                                        | 5%          | WB          | 4.7   | 5.6   | 9.9   | 14.0  | 20.0  | 22.8  | 24.1  | 23.5  | 21.8  | 17.8  | 12.4  | 7.3   | (31) |
|                                                                        |             | MCDB        | 5.7   | 7.4   | 13.1  | 18.1  | 24.1  | 27.8  | 28.1  | 27.2  | 25.2  | 20.4  | 14.1  | 8.6   | (32) |
|                                                                        | 10%         | WB          | 2.5   | 3.4   | 6.8   | 11.7  | 18.2  | 21.8  | 23.0  | 22.5  | 20.7  | 16.1  | 10.1  | 4.9   | (33) |
|                                                                        |             | MCDB        | 3.5   | 5.0   | 8.9   | 15.6  | 21.8  | 26.3  | 27.0  | 26.2  | 23.5  | 18.4  | 12.1  | 6.2   | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 7.7   | 8.7   | 9.5   | 10.9  | 11.5  | 11.2  | 10.9  | 10.2  | 11.0  | 9.8   | 7.8   | 6.9   | (35) |
|                                                                        |             | MCDBR       | 9.6   | 11.8  | 14.8  | 15.9  | 14.1  | 13.4  | 11.8  | 11.6  | 12.7  | 12.2  | 10.3  | 9.7   | (36) |
|                                                                        | 5% WB       | MCWBR       | 8.6   | 9.3   | 10.2  | 10.2  | 8.1   | 6.4   | 5.6   | 5.7   | 6.3   | 7.6   | 7.9   | 8.2   | (37) |
|                                                                        |             | MCWBR       | 9.1   | 11.3  | 14.0  | 13.8  | 12.6  | 11.9  | 10.5  | 10.2  | 10.8  | 10.5  | 9.4   | 9.2   | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.306 | 0.333 | 0.350 | 0.396 | 0.424 | 0.444 | 0.433 | 0.423 | 0.387 | 0.360 | 0.321 | 0.304 | (40) |
|                                                                        |             | taud        | 2.374 | 2.283 | 2.354 | 2.246 | 2.221 | 2.202 | 2.256 | 2.273 | 2.356 | 2.415 | 2.486 | 2.430 | (41) |
|                                                                        | Ebn at Noon | Ebn at Noon | 829   | 859   | 892   | 870   | 853   | 834   | 838   | 835   | 841   | 822   | 811   | 800   | (42) |
|                                                                        |             | Edh at Noon | 80    | 104   | 110   | 132   | 139   | 142   | 133   | 127   | 109   | 91    | 72    | 70    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.41  | 2.30  | 3.50  | 4.62  | 5.41  | 6.13  | 6.09  | 5.28  | 4.25  | 2.75  | 1.69  | 1.19  | (44) |
|                                                                        | RadStd      | RadStd      | 0.16  | 0.25  | 0.35  | 0.41  | 0.46  | 0.37  | 0.30  | 0.22  | 0.37  | 0.24  | 0.21  | 0.10  | (45) |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (58 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page